📘 MODULE 15: TSTATS

15.1 Concept

tstats runs statistics **directly on Splunk's index files (tsidx)** instead of reading raw events. It is dramatically faster than stats, but limited to certain fields.

15.2 Why it is fast

When Splunk indexes data, it writes two things per bucket:

- The **raw data** (compressed rawdata journal)
- A **tsidx index file**, which is like a book's index: a lookup from terms and **indexed fields** (host, source, sourcetype, index, _time) to the events that contain them.

stats must open and parse raw events, then extract fields at search time. tstats answers from the **tsidx** alone, with no raw-event decompression and no search-time extraction.

stats : disk → read raw events → extract fields → aggregate (slow)

tstats : disk → read tsidx index → aggregate (fast)

15.3 What tstats can use

1. **Indexed fields:** index, host, source, sourcetype, _time.
2. **Accelerated data models** (Module 16), which is the big use case.
3. Fields you explicitly indexed at ingest time (rare).

It **cannot** use search-time extracted fields (rex, user from auto-extraction) or free-text keywords in _raw.

15.4 Syntax

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| tstats <aggregate functions> [FROM <datamodel>] [WHERE <conditions>] [BY <fields> | _time span=<x>]

- It is a **generating command**, so it starts with |.
- WHERE index=... is where you filter, and you should always specify the index.
- FROM datamodel=<name> queries a data model.
- summariesonly=true queries **only accelerated summaries** (fast, but misses unaccelerated data). false also scans raw data for unsummarized parts.

15.5 Worked queries

Query 1: events by sourcetype

spl

```
| tstats count WHERE index=_internal BY sourcetype
```

What it does: counts events per sourcetype.
Compare: index=_internal | stats count by sourcetype. Same answer, much faster on big data.
Expected result: sourcetype | count.
Common mistake: missing leading pipe, or using lowercase-unfriendly typos (WHERE is conventionally uppercase).

Query 2: events over time

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```
| tstats count WHERE index=_internal BY _time span=1h
```

What it does: hourly event volume. Good for ingestion-volume dashboards.

Query 3: silent hosts (SOC and admin favorite)

spl

```
| tstats latest(_time) AS last_seen WHERE index=* BY host
| eval minutes_silent=round((now()-last_seen)/60)
| where minutes_silent>60
| eval last_seen=strftime(last_seen,"%F %T")
```

What it does: lists hosts that haven't sent any data for over an hour.
Interview explanation: "I use tstats on host and _time because it's index-level metadata, so this runs fast even across large indexes."

Query 4: data model (preview of Module 16)

spl

```
| tstats summariesonly=true count FROM datamodel=Authentication
  WHERE Authentication.action="failure"
  BY Authentication.src, Authentication.user
| rename Authentication.* AS *
```

What it does: counts failed authentications from the CIM Authentication data model.
Line-by-line: FROM datamodel= picks the data model · fields are **prefixed** with the dataset name · rename removes the prefix · summariesonly=true uses only accelerated data.

Requires: a CIM-compliant add-on, the CIM app, and acceleration enabled. It won't return data on a lab without these.

Example that fails

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| tstats count WHERE index=main BY user

user is extracted at search time, not indexed, so you'll get no useful result. Use stats, or map your data to a data model.

15.6 tstats vs stats

	stats	**tstats**
Reads	Raw events	Index (tsidx) files
Speed	Slower	Very fast
Fields available	Any (search-time too)	Indexed fields and accelerated data model fields
Starts with	After a search	| (generating command)
Free-text keyword filter	Yes	Not generally
Best for	Flexible analysis, small/medium data	Large-scale counts, volume, data model detections

15.7 When to use / not use

Use it for: event counts by host/sourcetype/index/time, ingestion monitoring, silent-host detection, data-model-based detections and dashboards.
Don't use it for: custom fields from rex, keyword searches in raw text, ad-hoc investigation needing full event detail.

15.8 Hands-on

1. Run | tstats count WHERE index=_internal BY sourcetype and compare run time with the stats version (check the job inspector).
2. Make an hourly event-volume tstats for your linux_auth index.
3. Build the "silent hosts" detection with a 10-minute threshold.
4. Predict what | tstats count WHERE index=_internal BY component returns, then run it. (Result: component is search-time, so expect nothing useful.)

15.9 Troubleshooting

Symptom	Cause
No results	Field isn't indexed, missing WHERE index=, or no accelerated data model
Data model query empty	Acceleration disabled, or CIM mapping missing (the data isn't tagged correctly)
Missing recent events	summariesonly=true and acceleration hasn't caught up
Syntax error	Missing leading |, or data model fields without the dataset prefix

15.10 Interview Q&A

- **Why is tstats fast?** It reads tsidx index files and doesn't touch raw events or do search-time extraction.
- **What fields can it use?** Indexed fields and accelerated data model fields.
- **tstats vs stats?** stats works on raw events with any field, tstats is much faster but limited.
- **What does summariesonly=true do?** Restricts the search to accelerated summaries.
- **Give a use case.** Finding silent hosts, or counting authentication failures via the Authentication data model.

15.11 Revision

tstats = reads the index, not events · generating command (|) · indexed fields + data models · summariesonly · fast but not flexible.